Certified Equilibrium Propagation in Non-Conservative Multistable Systems: Exact Gradient Conditions and Finite Perturbation Guarantees

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Abstract

Equilibrium propagation (EP) computes a learning signal from the response of a physical or simulated dynamical system to a weak perturbation of its stationary state. The classical guarantee rests on an energy function, which forces the state Jacobian to be symmetric. Recent work has removed the symmetry requirement, and finite-nudge exactness has been established for contrastive free-energy objectives in thermal ensembles. What remains open is computational: a finite-precision procedure returns two numerical states, and no criterion determines whether the gradient read from them meets a stated tolerance, or whether the states belong to the intended equilibrium branch of a multistable system. We address that question analytically and numerically. First, we isolate the exact gradient conditions for general smooth non-conservative equilibrium systems and show how the equilibrium sensitivity $A^{-1}q$ differs from the adjoint quantity $A^{-\top}q$ the parameter gradient requires, a distinction that vanishes in the conservative case and that explains the residual bias of standard vector-field EP. Second, we derive a finite-perturbation error budget with computable constants: a second-order contrast bias, a solver term that grows as $1/\beta_{\max}$, a finite-time integration term, and a rounding term. Third, we convert the budget into a fail-closed certification algorithm built on interval Krawczyk operators over exact rational arithmetic, which certifies branch existence and uniqueness uniformly over a parameter interval, encloses the centred finite difference, and accepts or rejects an update against a prescribed tolerance. On an exactly solvable two-dimensional non-conservative system with two stable equilibria and a saddle, both stable branches and both perturbation endpoints are certified with a preconditioned contraction factor of 2249/13600; the controller accepts $\beta_{\max} = 0.005$ with a worst-branch gradient error bound of 7.53×10^{-6} at tolerance 10^{-5} and rejects every larger tested candidate, including a deliberately broad box that spans all three equilibria. On a three-state non-normal benchmark the certified residual run bounds the gradient error by 4.39×10^{-4} at tolerance 5×10^{-4} while the observed error is 1.08×10^{-5}. We also quantify why finite-time guarantees must not be read off the spectrum: systems with fixed eigenvalues can amplify the adjoint response by a factor of 15.5, and the corresponding Lyapunov envelope, while valid, exceeds the observed error by four orders of magnitude. All numbers in this paper are produced by the accompanying package in under a minute.

Keywords: equilibrium propagation; non-conservative dynamics; implicit gradient; finite perturbation; interval arithmetic; Krawczyk operator; certification; multistable systems.

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1 Introduction

Equilibrium propagation (EP) trains a dynamical system by comparing two of its stationary states [15]. In the free phase the system relaxes under the input alone; in the nudged phase the loss contributes a weak force that pushes the state toward a target. The difference between the two

states, read through local quantities, estimates the gradient of the loss with respect to the parameters. Because the two phases use the same dynamics, EP is a candidate learning rule for analog, neuromorphic and physical substrates, where a distinct backward pass is unavailable [6, 19, 4].

The classical derivation assumes an energy function E with $\dot{z} = -\nabla_z E$. Then the state Jacobian at a stationary point is symmetric, the state response and the adjoint response coincide, and the two-phase contrast reproduces the gradient. Symmetric coupling is rarely available in hardware: memristive crossbars, optical and mechanical networks, and feedforward architectures all realize non-reciprocal interactions. Two lines of work address this. The vector-field generalization replaces the energy by an arbitrary force field and keeps the two-phase rule, at the cost of a bias that grows with the antisymmetric part of the Jacobian [16, 8]. More recently, exact-gradient algorithms for non-conservative systems have been proposed: an augmented learning phase whose Jacobian is corrected by the antisymmetric part of the free-equilibrium Jacobian, and a variational construction on a doubled state space [17]. In parallel, EP has been placed on a statistical foundation. For real-valued Gibbs ensembles the contrast between the nudged and free Helmholtz free energies has an exact gradient at every finite temperature and every finite nudging strength, so that the finite-nudge bias of the deterministic rule is a consequence of taking the zero-temperature limit rather than of contrastive learning itself [9]. Quantum and thermal versions of the same response identities are developed in [10].

These results settle what the algorithm computes in exact arithmetic, to first order, or in the thermodynamic limit. They do not address the status of the numerical output of a run, which is floating-point states obtained from a root finder or an ODE integrator; the reported gradient is a quotient by a small number; and in a multistable system the two states may belong to different equilibrium branches. A small residual at the returned states is not a correctness argument: it bounds how well the iterate solves the equations the practitioner wrote down, not the distance to the true stationary state, and it says nothing about branch identity. The question we address is therefore the following.

> Given a smooth non-conservative equilibrium system with a regular local equilibrium, a finite nudging strength, and a target tolerance, when may the EP gradient estimate computed from numerical stationary states be declared trustworthy, and how much error must be attributed to it?

Answering it requires three objects that are usually kept apart: the exact gradient identity, the finite-perturbation error budget, and a branch-aware numerical certificate. This paper joins them, and verifies the join on systems whose exact answers are known in rational arithmetic.

1.1 Contributions

1. **Exact gradient conditions and the sensitivity-adjoint dichotomy.** For $\dot{z} = F(z, \theta)$ with a regular equilibrium we give the implicit gradient $\nabla_\theta L = \partial_\theta C - B^\top A^{-\top} q$ (Theorem 2), the exactness condition for standard vector-field EP as a single bias term $B^\top (A^{-1} - A^{-\top}) q$ together with a task-restricted exactness criterion (Theorem 4), and the observation that the centred contrast of the *uncorrected* nudged system measures the equilibrium sensitivity $A^{-1} q$ rather than the adjoint $A^{-\top} q$ (Proposition 3). The two coincide only when $A = A^\top$, which is why conservative EP cannot see the distinction and why non-conservative EP must.

2. **Finite-perturbation budget with computable constants.** We prove that the centred contrast of the antisymmetry-corrected learning phase is second order in the nudging strength (Theorem 9), identify the resulting trade-off between the $O(\beta_{\max}^2)$ contrast bias and the

$O(1/\beta_{\max})$ amplification of state error, give the corresponding optimal scale $\beta^* = (d/(2a))^{1/3}$, and assemble the budget $E_{\text{total}} \le E_\beta + E_z + E_t + E_{\text{fp}}$ with every constant obtained by interval evaluation rather than fitted.

3. **Branch-aware certification.** We certify existence and uniqueness of the free equilibrium, of both perturbation endpoints, and of the whole branch tube uniformly over a parameter interval, using exact rational Krawczyk operators, together with a Hurwitz test that is uniform on the tube; the centred finite difference is enclosed by interval arithmetic and compared with the exact rational gradient. The resulting algorithm is fail-closed: it either returns an interval containing the gradient with a certified error bound, or it refuses.

4. **Verified benchmarks with rejection paths.** A two-dimensional non-conservative system with two stable equilibria and a saddle is certified end to end in exact rational arithmetic, including the rejection of a wide box that contains all three equilibria and the rejection of nudging strengths whose bound exceeds tolerance. A three-state non-normal benchmark carries a certified residual run, an adaptive controller whose selected settings are cross-checked against a high-accuracy integrator, and a bounded-state-noise stress test.

1.2 Scope

We establish certified gradient conditions for a broad class of smooth non-conservative equilibrium systems with regular isolated equilibria ($F \in C^3$, $\sigma_{\min}(A) > 0$). We do not claim convergence of arbitrary nonlinear dynamics of arbitrary stability type, we do not claim global optimality of the selected nudging strength, and we do not claim that a certificate transfers to systems whose coefficients are uncertain or whose trajectories are only partially observed. The certified statements in Section 6 are statements about the explicitly defined benchmarks; their conclusions are intervals that contain the exact rational answer, not performance claims about neural networks. We do not propose a new EP update rule: the antisymmetry-corrected phase and the doubled-state construction are prior work [17, 4], and we use them as the objects of certification. The unit of certification is one gradient estimate, at one parameter point, on one certified branch: nothing here certifies the training trajectory that later consumes the estimate.

2 Related work

Equilibrium propagation and energy-based models. EP was introduced for continuous Hopfield energies and proved to perform gradient descent on a squared-error objective [15, 5]. Its second phase is equal, at every instant, to the error propagation of recurrent backpropagation [2, 13]. Contrastive Hebbian learning is the historical relative [11]; the equivalence of EP and contrastive rules for energy-based models is part of the original derivation.

Beyond symmetric coupling. The vector-field generalization removes the energy assumption and with it the symmetry of the Jacobian [16]. The two sources of bias, finite nudge and Jacobian asymmetry, were separated in [8], where a homeostatic penalty on the antisymmetric part of the Jacobian substantially improves performance without enforcing weight symmetry, and where exact derivatives are recovered in complex-differentiable networks by a Cauchy integral. An exact gradient for general non-conservative systems was obtained in [17] by adding a learning-phase force proportional to the antisymmetric part of the free-equilibrium Jacobian, equivalently by a variational construction on a doubled state space; dual propagation is the related compartmental construction [4]. Our certification applies to these constructions.

Finite nudging. Finite nudging is not a defect per se. For Gibbs ensembles the two-phase contrast is the exact gradient of a free-energy difference at any finite nudging strength and temperature, and a low-temperature limit recovers the deterministic rule [9]; the quantum and thermal cases are treated in [10]. The quantity that remains to be controlled in the deterministic setting is therefore the distance between the deterministic finite-nudge contrast and the exact gradient, which is the object of Theorem 9.

Implicit differentiation and stationary-state models. Gradients of a stationary state with respect to parameters are given by the implicit function theorem; deep equilibrium models make the same construction computational [1, 8]. Implicit differentiation provides the gradient analytically; the question we study is when a *dynamical* perturbation procedure realizes it with a finite error guarantee, which is a stricter requirement than differentiability.

Verified numerics. The Krawczyk operator is a classical inclusion test for systems of equations [7, 12]; verification methods based on floating-point interval arithmetic can produce rigorous inclusions at moderate cost [14]. We use them here over exact rational arithmetic rather than directed rounding, so that the accepted conclusions carry no rounding model beyond the explicitly converted binary rationals of the candidate states.

Non-normality. When the state Jacobian is far from symmetric, the spectrum does not control transient behaviour; the relevant objects are the numerical range, the semigroup norm and the pseudospectrum [18, 3]. Section 6.4 shows that this is not a technicality for EP: the finite-time adjoint tail is governed by the semigroup, not by the spectral abscissa.

3 Setting and notation

3.1 Parameterized equilibrium systems

Let $z \in \mathbb{R}^n$ denote the state and $\theta \in \mathbb{R}^p$ the learnable parameters of a dynamical system

$$\dot{z} = F(z, \theta), \qquad F \in C^3(\mathbb{R}^n \times \mathbb{R}^p; \mathbb{R}^n), \tag{1}$$

whose trajectories converge to a stationary state. No scalar potential is assumed: F need not equal $-\nabla_z E$ for any E, and the Jacobian

$$A = D_z F(z^\star, \theta), \qquad B = D_\theta F(z^\star, \theta) \tag{2}$$

at the free equilibrium $z^\star$ need not be symmetric. Let C be a differentiable loss and

$$q = \nabla_z C(z^\star, \theta) \tag{3}$$

its state gradient at the free equilibrium; C may depend explicitly on θ, in which case $\partial_\theta C$ denotes that partial derivative.

Assumption 1 (Regular local equilibrium)**.** $F(z^\star, \theta) = 0$ and $\det A \neq 0$, so that by the implicit function theorem a unique local branch $z^\star(\theta)$ of equilibria passes through $(z^\star, \theta)$.

Assumption 1 is the only regularity requirement behind the gradient identities of Section 4. Two further conditions appear later: a uniform contraction bound on a neighbourhood of the branch wherever we certify numerically, and a bounded certified parameter domain wherever the perturbation strength is treated as an interval.

3.2 Three learning-phase dynamics

The three dynamics below are the objects the paper distinguishes. All three read a gradient estimate out of a centred contrast of two stationary states,

$$\widehat{g}_\beta = B^\top \frac{z^{+\beta} - z^{-\beta}}{2\beta}, \tag{4}$$

and they differ only in the equations that define those states.

(i) Standard vector-field EP (uncorrected).

$$F(z^{\pm\beta}, \theta) \mp \beta \nabla_z C(z^{\pm\beta}, \theta) = 0. \tag{5}$$

(ii) Antisymmetry-corrected learning phase (AsymEP). With $A_{\mathrm{a}} = \frac{1}{2}(A - A^\top)$ evaluated at the free equilibrium and held fixed during the learning phase,

$$F(z^{\pm\beta}, \theta) \mp \beta \nabla_z C(z^{\pm\beta}, \theta) - 2A_{\mathrm{a}}\big(z^{\pm\beta} - z^\star\big) = 0. \tag{6}$$

(iii) Doubled-state system (Dyadic EP). With midpoint $m = \frac{1}{2}(z + z')$ and difference $d = z - z'$, the stationary state of the doubled system satisfies

$$F(m, \theta) = 0, \qquad A(m, \theta)^\top d = \beta \, \nabla_m C(m, \theta), \tag{7}$$

where $A(m, \theta) = D_m F(m, \theta)$ [17]. Equation (7) uses the centred contrast (4) in the form $\widehat{g}_\beta = B(m, \theta)^\top d/\beta$.

3.3 Three levels of evidence

Throughout, we separate three levels that are often conflated.

- **Analytic**: an identity, inequality or criterion proved for the model as written, without reference to a computation.
- **Floating-point diagnostic**: numbers produced by a numerical solver, informative about behaviour but not a proof of anything, in particular not of the distance to an exact root.
- **Certificate**: an enclosure produced by interval arithmetic from the exact rational conversion of the candidate states, whose acceptance is a mathematical statement about the model. A certificate can fail; failure is reported, never repaired by tightening a tolerance after the fact.

4 Exact gradient conditions

4.1 The implicit gradient

Theorem 2 (Equilibrium sensitivity and implicit gradient)**.** *Let Assumption 1 hold, let $L(\theta) = C(z^\star(\theta), \theta)$ be the objective evaluated at the equilibrium branch, and let A, B, q be as in (2)–(3). Then*

$$D_\theta z^\star = -A^{-1}B, \qquad \boxed{\nabla_\theta L = \partial_\theta C - B^\top A^{-\top} q} \tag{8}$$

and, when C does not depend explicitly on θ, the true negative gradient reads $d^\star = B^\top A^{-\top} q$.

The proof is in Appendix A. Equation (8) requires nothing beyond a regular local equilibrium; it applies to dissipative, non-reciprocal and feedforward force fields alike. It is also the exact target that any EP procedure has to approximate, and every error bound below is measured against it.

4.2 Sensitivity is not the gradient

Proposition 3 (Sensitivity and adjoint)**.** *Define the equilibrium sensitivity s and the adjoint λ by*

$$As = q, \qquad A^\top \lambda = q, \tag{9}$$

that is $s = A^{-1}q$ and $\lambda = A^{-\top}q$. The centred contrast of the uncorrected nudged system (5) satisfies

$$\frac{z^{+\beta} - z^{-\beta}}{2\beta} = s + O(\beta^2), \tag{10}$$

while the parameter gradient of Theorem 2 is read through λ. If $A = A^\top$ then $s = \lambda$ and the two notions coincide.

Proposition 3 is the reason non-conservative systems need a corrected learning phase. It also identifies a failure mode of the numerical practice: a two-phase run with small residuals may still yield an incorrect direction whenever A is not symmetric, and the discrepancy is not small unless q happens to lie in a subspace on which the two responses agree. Table 1 records this on the benchmark of Section 6.1: the two quantities are -0.3235294118 and -0.3627450980, a relative difference of 12%.

4.3 Bias of the uncorrected vector-field rule

Theorem 4 (Vector-field EP bias)**.** *Let the uncorrected states solve (5) and let $\widehat{g}_\beta$ be the contrast (4). Then the infinitesimal contrast is $B^\top A^{-1} q$, so that against the true negative gradient $B^\top A^{-\top} q$ the bias is*

$$\boxed{e_{\mathrm{VF}} = B^\top \big(A^{-1} - A^{-\top}\big) q} \tag{11}$$

and the rule is exact at the given parameter point for the given task if and only if $B^\top (A^{-1} - A^{-\top}) q = 0$. If the loss gradients range over a subspace $S \subset \mathbb{R}^n$, exactness for every task in S holds if and only if

$$B^\top \big(A^{-1} - A^{-\top}\big) S = \{0\}. \tag{12}$$

Example 5 (Task-restricted exactness without symmetry)**.** Take

$$A = \begin{pmatrix} -1 & 6 \\ 0 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad S = \mathrm{span}\{(1,0)^\top\}. \tag{13}$$

Then A is not symmetric, $B^\top (A^{-1} - A^{-\top}) = (0, -6)$, the operator norm of (12) restricted to S is 0, and the same operator has norm 6 over all unit loss gradients. For $q = (1,0)^\top$ the uncorrected contrast is exact; for $q = (0,1)^\top$ its error is 6. Stability of the equilibrium does not remove the bias.

Example 5 is the sharpest form of the statement that symmetry is not needed task by task, only in the directions the task actually excites. It is reproduced by `run_stage2.py`.

4.4 Antisymmetry correction and the adjoint

Theorem 6 (Corrected learning phase realises the adjoint)**.** *Let $A_{\mathrm{a}} = \frac{1}{2}(A - A^\top)$ and let $g(z, \beta) = F(z, \theta) - 2A_{\mathrm{a}}(z - z^\star) - \beta \nabla_z C(z, \theta)$ be the corrected learning field (6). Then*

$$D_z g(z^\star, 0) = A^\top, \qquad \textit{hence} \qquad \frac{\mathrm{d}z^\beta}{\mathrm{d}\beta}\Big|_{\beta=0} = A^{-\top} q = \lambda. \tag{14}$$

Consequently the infinitesimal contrast of the corrected phase equals the adjoint, and the residual error at finite β is the second-order term quantified in Theorem 9.

Table 1: Evidence levels for the same benchmark branch $z^\star = (1, 0)$ of the system (25). The implicit gradient is the analytic target; the uncorrected rule is biased at any nudging strength; the corrected finite-nudge contrast converges quadratically; the doubled state identity is exact in algebra and certified in finite precision by Theorem 8. Errors are absolute; the exact gradient is $-11/34$.

Method	Compared quantity	Representative result	Evidence level
Implicit gradient	$B^\top A^{-\top} q$	-0.3235294118	analytic
Uncorrected vector-field EP	$B^\top A^{-1} q$	bias 3.9216×10^{-2}	analytic
Corrected contrast, $\beta = 0.01$	centred finite difference	error 3.0732×10^{-6}	floating point
Corrected contrast, $\beta = 0.005$	centred finite difference	error 7.6826×10^{-7}	floating point
Doubled state, ideal stationarity	d/β	error 0	algebraic identity
Doubled state, float64 solve	exact residual recomputation	error $\leq 1.1973 \times 10^{-16}$	rational certificate

4.5 Finite nudging with an exact midpoint

Theorem 7 (Dyadic stationarity gives the adjoint exactly)**.** *Suppose the doubled system (7) reaches a stationary state whose midpoint is the free equilibrium, $F(m, \theta) = 0$, and whose difference satisfies $A(m, \theta)^\top d = \beta \nabla_m C(m, \theta)$ with $\beta \neq 0$. If $A(m, \theta)$ is nonsingular then*

$$\boxed{\frac{d}{\beta} = A(m, \theta)^{-\top} \nabla_m C(m, \theta)} \tag{15}$$

at that finite β; there is no limiting argument. If moreover the midpoint is the exact free equilibrium and C does not depend explicitly on θ, then $B^\top d/\beta$ equals the true negative gradient of Theorem 2.

Theorem 7 is an exactness statement about the stationary equations, not about an algorithm: it is silent about how the stationary state is reached, how far a returned iterate is from it, and whether the midpoint is the intended branch. The next theorem converts the residual of the returned iterate into a gradient error.

Theorem 8 (Residual-to-gradient transfer)**.** *Let $(\widehat{m}, \widehat{d})$ be any numerical candidate with $A(\widehat{m})$ nonsingular, and define the difference-equation residual $r_d = A(\widehat{m})^\top \widehat{d} - \beta \nabla C(\widehat{m})$. Then*

$$\left\| \frac{\widehat{d}}{\beta} - A(\widehat{m})^{-\top} \nabla C(\widehat{m}) \right\| \leq \frac{\|A(\widehat{m})^{-\top}\| \, \|r_d\|}{|\beta|}, \tag{16}$$

and on the benchmark of Section 6.1 the exact constant is $\|A^{-\top}\|_\infty = 55/51$, evaluated in rational arithmetic.

Equation (16) contains the mechanism that makes finite-precision certification necessary: the residual is divided by $|\beta|$, so decreasing the nudging strength to reduce contrast bias amplifies solver error. Section 5 quantifies the balance.

5 Finite-perturbation analysis and certification

5.1 Second-order contrast and the nudging trade-off

Theorem 9 (Centred contrast is second order)**.** *Let $z^\star$ satisfy Assumption 1, and for $|\beta| \leq \beta_{\max}$ let $\varphi(\beta)$ be the corrected learning-phase root of (6) near $z^\star$, so that φ is C^3 and $\varphi(0) = z^\star$ with*

$\varphi'(0) = \lambda = A^{-\top} q$. *Then*

$$\Big\| \frac{\varphi(\beta) - \varphi(-\beta)}{2\beta} - \lambda \Big\| \;\le\; \frac{M_3}{6}\beta^2, \qquad M_3 = \sup_{|\beta'| \le \beta_{\max}} \|\varphi'''(\beta')\|, \tag{17}$$

and consequently, with $\widehat{g}_\beta$ *from* (4),

$$\|\widehat{g}_\beta - d^\star\| \;\le\; \frac{\|B\|}{6} M_3 \beta^2 + \textit{(solver and integration terms).} \tag{18}$$

The odd-order term cancels in the centred difference; this is the deterministic counterpart of the finite-nudge exactness obtained for free-energy contrasts in thermal ensembles [9], where the residual is a zero-temperature effect rather than a contrastive one.

Theorem 9 replaces the empirical observation that the error is approximately quadratic with a bound whose constant is computed, not fitted. In the implementation of Section 6.4 the constant is obtained by differentiating the stationarity equation implicitly: with a certified contraction rate α on a neighbourhood of the branch, one obtains upper bounds $\|\varphi'\| \le x_1$, $\|\varphi''\| \le x_2$, $\|\varphi'''\| \le x_3$ from the derivative bounds on F and ∇C, and

$$x_1 \le \sup_{|\beta'| \le \beta_{\max}} \|\varphi'(\beta')\|, \qquad x_2 \le \sup_{|\beta'| \le \beta_{\max}} \|\varphi''(\beta')\|, \qquad x_3 \le \sup_{|\beta'| \le \beta_{\max}} \|\varphi'''(\beta')\|, \qquad E_\beta = \frac{\|B\|\,|\beta|^2 x_3}{6}. \tag{19}$$

For the three-state benchmark this gives $E_\beta = 3.656 \times 10^{-4}$ at $\beta = 0.005$; the same expression evaluates to 6.07×10^{-3} and 1.48×10^{-3} at $\beta = 0.02$ and 0.01, which is why those candidates are discarded before any root finding is attempted.

Proposition 10 (Error balance and an optimal scale)**.** *Suppose the contrast bias is bounded by* $E_\beta = a\beta^2$ *and the state-error contribution by* $E_z = d/\beta$ *with* $a, d > 0$ *independent of* β*. Then the total bound* $a\beta^2 + d/\beta$ *is minimised at*

$$\beta^* = \Big(\frac{d}{2a}\Big)^{1/3}, \tag{20}$$

and it is not monotone in β*: decreasing the nudging strength indefinitely increases the bound. On the three-state benchmark (*$a = 0.1208$*,* $d = 2.94 \times 10^{-4}$ *per phase) this gives* $\beta^* = 0.107$ *for the stated budget, against an empirical best value* 0.085 *on the scanned grid; the difference is the gap between an upper-bound proxy and measured error.*

5.2 The total budget

Theorem 11 (Certified finite-perturbation bound)**.** *Let Assumption 1 hold on a neighbourhood* Ω *of the branch with* $\sigma_{\min}(A(z)) \ge \alpha > 0$ *for* $z \in \Omega$*, let* $|\beta| \le \beta_{\max}$ *be such that the perturbation endpoints lie in* Ω*, and let* $(\widehat{z}^{+\beta}, \widehat{z}^{-\beta})$ *be computed states with certified boxes* X^+*,* X^- *that contain both the exact roots* $z^{+\beta}, z^{-\beta}$ *and the reported candidates* $\widehat{z}^{+\beta}, \widehat{z}^{-\beta}$*, and, when the states are obtained by integrating an auxiliary dynamics for time* T*, a certified finite-time error* δ_t*. Then*

$$\boxed{\|\widehat{g}_\beta - d^\star\| \;\le\; \underbrace{\frac{\|B\|}{6} M_3 \beta^2}_{E_\beta:\ \textit{contrast bias}} + \underbrace{\frac{\|B\|}{4\beta_{\max}}\big(\operatorname{diam} X^+ + \operatorname{diam} X^-\big)}_{E_z:\ \textit{root enclosure}} + \underbrace{\frac{\|B\|}{\beta_{\max}}\delta_t}_{E_t:\ \textit{finite time}} + \underbrace{\kappa(A)\,u}_{E_{\mathrm{fp}}:\ \textit{rounding}}} \tag{21}$$

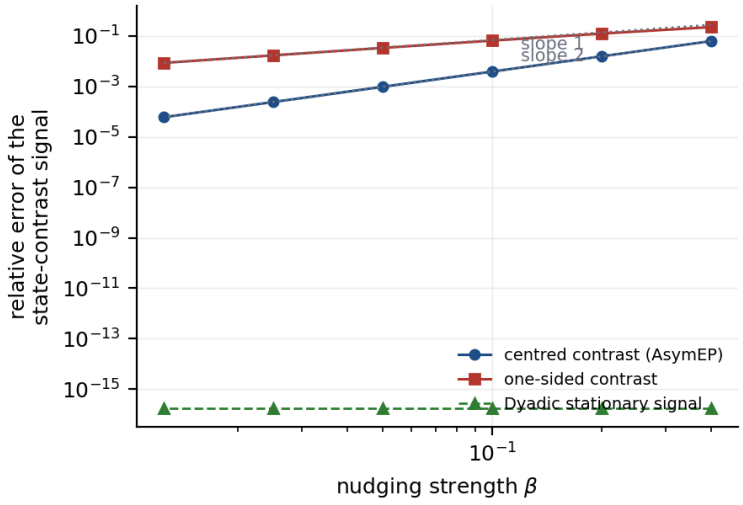


Figure 1: Relative error of the state-contrast signal against the nudging strength on the three-state benchmark of Section 6.4, log-log scale. The centred contrast of the corrected phase falls with slope 2.002, the one-sided contrast with slope 0.946, and the doubled-state stationary signal sits at machine precision. The fitted orders are produced by the same run that produces the curves.

with u the unit roundoff and $\kappa(A) = \|A\|\|A^{-1}\|$. Each term is either certified by interval evaluation or explicitly declared as a modelling assumption; none is fitted. The root-enclosure term follows from the estimator $\widehat{g}_\beta = B^\top(z^{+\beta} - z^{-\beta})/(2\beta)$ together with $\operatorname{diam} X^\pm = 2r^\pm$ for boxes of radius $r^\pm$.

Theorem 11 is the analytic counterpart of Algorithm 1. Four remarks are in order. First, the budget is stated for $\beta_{\max}$, the largest perturbation strength in the certified domain, so that a single tube certificate covers every candidate below it. Second, a bound of the form (21) is only meaningful when each term is finite on the certified domain: the contraction certificate below is what makes that true. Third, the verdict is a comparison of the bound with a tolerance, not a comparison of the observed error with the tolerance; the latter would be circular, since the observed error is what the certificate is supposed to bound. Fourth, the four terms are alternatives rather than a checklist that a run must populate. On the benchmarks of Section 6 the finite-time term is realised in a stronger form than the statement requires: instead of bounding an integrator's truncation error, the endpoint certificate bounds the residual of the returned state and converts it into a distance to the equilibrium through the certified contraction rate, so E_t is replaced by an E_z computed from that distance, not by an error estimate for the integration. The rounding term E_{fp} is absent for a different reason: candidate states are converted to exact binary rationals and every subsequent operation is exact, so no rounding enters those certificates at all and $\kappa(A)u$ overstates a contribution that is zero in that arithmetic.

Two routes to a gradient bound are available, and the two benchmarks exercise different routes. When the exact rational gradient $d^\star$ of Theorem 2 has a closed form, the verdict uses the direct interval error

$$D(\beta) = [\underline{d},\ \overline{d}] \supseteq \widehat{g}_\beta, \qquad E(\beta) = \max\bigl\{|\underline{d} - d^\star|,\ |\overline{d} - d^\star|\bigr\}, \tag{22}$$

so that $|\widehat{g}_\beta - d^\star| \le E(\beta)$ holds by construction; this is the quantity reported in Tables 2 and 3. It covers both the contrast bias and the enclosure radius of the endpoints, and it requires no Taylor remainder estimate, since the truth is known exactly. It is nevertheless a statement about the

estimator at one parameter point rather than a model-independent guarantee, because it presupposes the closed-form truth; that is precisely why it is used only on the rational benchmark. The truth-free route is (21), whose terms are computed by interval evaluation alone: it is the route the three-state benchmark of Section 6.4 takes, and the route available to a user who has no closed-form gradient, at the cost of constants that may be conservative by orders of magnitude.

5.3 Branch-existence certificates and the fail-closed rule

Theorem 12 (Krawczyk inclusion). *Let $g : \mathbb{R}^n \to \mathbb{R}^n$ be C^1 on a box X, let $x \in X$, let P be a nonsingular $n \times n$ matrix, and let $Dg(X)$ be an interval enclosure of the Jacobian on X. Define*

$$K(X) = x - P\,g(x) + \big(I - P\,Dg(X)\big)(X - x). \tag{23}$$

If $K(X) \subset \mathrm{int}(X)$ and $\|I - P\,Dg(X)\|_\infty < 1$, then g has exactly one zero in X, and the preconditioned fixed-point map is a contraction on X.

Corollary 13 (Uniform certification over a parameter interval). *Let $g(z, \beta)$ depend on a parameter $\beta \in \mathcal{B}$ and suppose that for the same box X and preconditioner P the inclusion $K(X, \mathcal{B}) \subset \mathrm{int}(X)$ holds with an interval Jacobian evaluated over both X and $\mathcal{B}$, and that $\sup_{\beta\in\mathcal{B}}\|I - PD_z g(X, \beta)\|_\infty = c < 1$. Then for every fixed $\beta \in \mathcal{B}$ there is exactly one root in X. If in addition the trace and determinant of the learning-phase Jacobian $D_z g$ are sign-definite on $X \times \mathcal{B}$ (negative trace, positive determinant in the two-dimensional case), that root is locally asymptotically stable as a state of the learning-phase dynamics. Stability of the free equilibrium is a separate property, governed by $A = D_z F$ and certified separately.*

Corollary 13 is the reason a certification run can cover a continuum of nudging strengths rather than a single value, and it is the device that lets a controller search over candidate strengths without re-certifying existence. Branch identity is then a consequence of geometry rather than of the solver's initial guess: a box whose solution set is certified unique cannot contain two distinct equilibria, so a deliberately wide box spanning several equilibria must be rejected by the inclusion test. That rejection is itself part of the acceptance suite. Containment carries the same argument one level further. Since the box holds exactly one root for each β, an endpoint verified to lie inside the tube box lies on the same branch as the free equilibrium, and the two endpoints of a centred contrast belong to one branch rather than to two. Without that containment check the two certified endpoints could sit on two different branches and their difference quotient would be meaningless, which is the failure mode the final rejection rule of Algorithm 1 detects.

Theorem 14 (Local uniqueness is all that is claimed). *The conclusion of Theorem 12 and Corollary 13 is local: it asserts existence and uniqueness in the certified box, and no global statement about the number of equilibria or about convergence of trajectories. Consequently a certificate can be produced for one branch while another branch, equally stable, is not certified in the same run; and the presence of an uncertified branch is not evidence against the certified one. The framework is sound but not complete: a failed inclusion says that the box was too wide for the test, not that no root is there, so a rejection is a statement about the certificate and never a claim about the system.*

5.4 Non-normality: finite-time bounds are not spectral bounds

Theorem 15 (Transient amplification). *Let $A_K = \begin{pmatrix} -1 & K \\ 0 & -1 \end{pmatrix}$ with $K \geq 0$. All eigenvalues equal -1, so the spectral abscissa is -1 for every K, yet*

$$\|e^{A_K^\top t}\|_2 = e^{-t}\sqrt{1 + \tfrac{K^2t^2}{2} + \tfrac{Kt}{2}\sqrt{K^2t^2 + 4}} \tag{24}$$

Algorithm 1 Certified finite-perturbation gradient update (fail-closed)

1. **Free equilibrium.** Obtain a candidate $z^{(0)}$ by Newton iteration on $F(z,\theta)=0$; build a rational box $X^\star$ around the exact rational conversion of $z^{(0)}$; evaluate (23) for $g=F$. If the inclusion fails, halve the box and retry up to a fixed budget; if it still fails, REJECT.

2. **Branch tube.** For the candidate strength $\beta_{\max}$, build the corrected learning field $g(z,\beta)=F(z,\theta)-2A_{\mathrm{a}}(z-z^\star)-\beta\nabla_z C(z,\theta)$ and certify $K(X,\mathcal{B})\subset\mathrm{int}(X)$ over the parameter interval $\mathcal{B}=[-\beta_{\max},\beta_{\max}]$ with the contraction factor of Corollary 13; certify the Hurwitz signs of the learning Jacobian on the same box.

3. **Endpoints.** For each of $\pm\beta$, run a floating-point root solve, convert the returned coordinates to exact binary rationals, build a small rational box and certify it by Theorem 12. Check that both endpoint boxes lie inside the certified tube box; otherwise REJECT, since the two states are then not certified to belong to the same branch.

4. **Gradient enclosure.** Form the interval $D=(X^+-X^-)/(2\beta)$ and the error bound $E(\beta)=\max\{|\underline{d}-d^\star|,|\overline{d}-d^\star|\}$ where $\underline{d},\overline{d}$ are its endpoints and $d^\star$ is the exact rational implicit gradient (Theorem 2); in the absence of a closed-form $d^\star$, use (21).

5. **Verdict.** If all certificates pass and $E(\beta)\le\varepsilon$, return the interval $[\underline{d},\overline{d}]$ with its certificate. Otherwise move to the next candidate β in descending order; if no candidate passes, REJECT and report which obligation failed.

which grows before it decays. Two families appear below and should not be conflated. The closed form (24) is for the equal-eigenvalue case, whose two eigenvalues are both -1; it is the family used for the counterexample of Section 6.4. The scanned family $A_K=\begin{pmatrix}-1 & K\\ 0 & -0.1\end{pmatrix}$, *with eigenvalues* -1 *and* -0.1, *has no equally compact form and is evaluated numerically; the same phenomenon appears there, with the semigroup norm reaching* 15.5 *at* $K=20$. *In both families the Lyapunov envelope* $\|e^{A^\top t}\|_2\le\sqrt{\kappa(P)}\,e^{-\lambda t}$ *with* $AP+PA^\top=-I$ *and* $\lambda=1/(2\|P\|)$ *remains valid, but it is not tight: on the scanned family it stays above the directional adjoint error by four orders of magnitude while never being violated, and the direct operator-norm bound is itself three orders above the error. Therefore the finite-time term* E_t *of Theorem 11 must be derived from a semigroup bound or a certified integrator estimate, not from the spectrum, and a stability certificate for the equilibrium does not by itself justify a finite-time error bound.*

6 Numerical verification

Every number below is produced by the accompanying package. The package runs nine stages and an automated test suite; a single command reproduces all tables, figures and certificates, and each stage exits non-zero if one of its assertions fails. Certificates that are accepted were produced by exact rational interval arithmetic over `fractions.Fraction`; floating-point root finders supply candidates only.

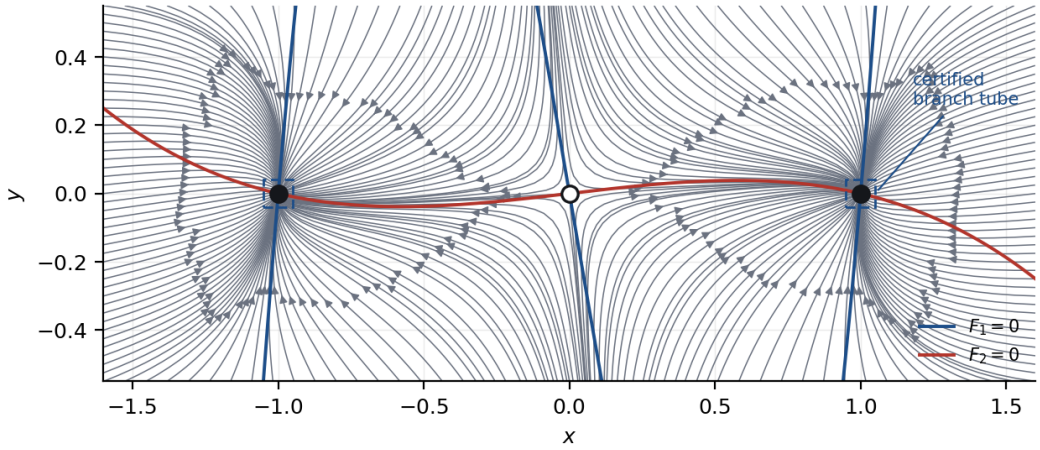


Figure 2: The benchmark (25). Nullclines $F_1 = 0$ and $F_2 = 0$, the flow field, and the three equilibria (filled: stable, open: saddle). Dashed rectangles are the branch tubes certified in Section 6.1: the Krawczyk inclusion holds uniformly for every $\beta \in [-0.02, 0.02]$ on both tubes.

6.1 An exactly solvable non-conservative system with two stable branches

Take

$$F(x, y; \theta) = \begin{pmatrix} x - x^3 + \frac{1}{5}y + \theta \\ -y + \frac{1}{10}(x - x^3) \end{pmatrix}, \qquad C(x, y) = \tfrac{1}{2}\Big[\big(x - \tfrac{3}{10}\big)^2 + \big(y + \tfrac{1}{5}\big)^2\Big], \tag{25}$$

with $\theta = 0$. The second equation gives $y = (x - x^3)/10$; substituted into the first it yields $\frac{51}{50}x(1 - x^2) = 0$, so the equilibria are exactly

$$z_L = (-1, 0), \qquad z_s = (0, 0), \qquad z_R = (1, 0), \tag{26}$$

and the state Jacobian is

$$A(x) = \begin{pmatrix} 1 - 3x^2 & \frac{1}{5} \\ \frac{1}{10}(1 - 3x^2) & -1 \end{pmatrix}. \tag{27}$$

At $x = \pm 1$ we have $\operatorname{tr} A = -3$ and $\det A = 51/25 > 0$, so both outer equilibria are locally asymptotically stable; at the origin $\det A = -51/50 < 0$, so it is a saddle. The two off-diagonal entries are $1/5$ and $-1/5$, so A is not symmetric: the benchmark is non-conservative and multistable at the same time, and the global uniqueness of equilibria required by conservative arguments fails (Figure 2). The parameter enters only the first equation additively, hence $B = (1, 0)^\top$, and the exact negative gradients are

$$d^\star(z_R) = -\frac{11}{34} \approx -0.3235294118, \qquad d^\star(z_L) = \frac{67}{102} \approx 0.6568627451, \tag{28}$$

computed from Theorem 2 in rational arithmetic, not estimated by finite differences.

Branch tubes. For each stable branch we take the box $X_s = [s - 0.05, s + 0.05] \times [-0.04, 0.04]$ and the parameter interval $\mathcal{B} = [-0.02, 0.02]$, and evaluate the parametric Krawczyk operator (23) in exact rational arithmetic with the preconditioner $P = \big(D_z g(z_s, 0)\big)^{-1} = \big(A_s^\top\big)^{-1} = A_s^{-\top}$. Both tubes pass the strict inclusion test with the same certified contraction factor

$$\sup_{\beta \in \mathcal{B}} \| I - P\, D_z g(X_s, \beta) \|_\infty = \frac{2249}{13600} \approx 0.165368 < 1, \tag{29}$$

Table 2: Certified worst-branch gradient error bound against the nudging strength on the benchmark (25), tolerance $\varepsilon = 10^{-5}$. Every row is the maximum over the two stable branches of the interval error (22). The bound scales as β^2: the ratios for successive halvings are 4.0008, 4.0011, 4.0007. Acceptance stops between 0.0075 and 0.005.

β	worst-branch error bound	branch bounds $(-1,0)$ / $(1,0)$	decision
0.02000	1.2019×10^{-4}	1.2019×10^{-4} / 1.2588×10^{-5}	rejected, exceeds tolerance
0.01500	6.7593×10^{-5}	6.7593×10^{-5} / 7.0815×10^{-6}	rejected, exceeds tolerance
0.01250	4.6938×10^{-5}	4.6938×10^{-5} / 4.9174×10^{-6}	rejected, exceeds tolerance
0.01000	3.0042×10^{-5}	3.0042×10^{-5} / 3.1473×10^{-6}	rejected, exceeds tolerance
0.00750	1.6905×10^{-5}	1.6905×10^{-5} / 1.7708×10^{-6}	rejected, exceeds tolerance
0.00500	$\mathbf{7.5271 \times 10^{-6}}$	7.5271×10^{-6} / 7.8826×10^{-7}	**accepted**
0.00400	4.8295×10^{-6}	4.8295×10^{-6} / 5.0570×10^{-7}	accepted
0.00250	1.9167×10^{-6}	1.9167×10^{-6} / 2.0069×10^{-7}	accepted
0.00125	5.4918×10^{-7}	5.4918×10^{-7} / 5.7493×10^{-8}	accepted
0.00100	4.0027×10^{-7}	4.0027×10^{-7} / 4.1899×10^{-8}	accepted

Table 3: Branch-level certificates at $\beta = 0.005$. The interval column is the certified enclosure of the centred finite difference (4); the error bound is (22) against the exact rational gradient; the observed error is the floating-point finite difference evaluated at the returned interior point. All four endpoint boxes are certified by Theorem 12 and lie strictly inside the certified tube boxes.

Stable branch	exact $d^\star$	certified difference interval	certified bound	observed error
$(-1,0)$	$67/102 = 0.6568627451$	$[0.6568702322,\ 0.6568702722]$	7.5271×10^{-6}	7.5071×10^{-6}
$(1,0)$	$-11/34 = -0.3235294118$	$[-0.3235286635,\ -0.3235286235]$	7.8826×10^{-7}	7.6826×10^{-7}

so by Corollary 13 each tube contains exactly one learning-phase root for every β in the interval. On the same tube the learning Jacobian satisfies

$$\operatorname{tr} D_z g \in \left[-\tfrac{1339}{400}, -\tfrac{1067}{400}\right] = [-3.3475, -2.6675], \qquad \det D_z g \in \left[\tfrac{4219}{2500}, \tfrac{24199}{10000}\right] = [1.6876, 2.4199], \tag{30}$$

so the Hurwitz criterion holds throughout the tube: every learning-phase root in it is locally asymptotically stable as a state of the augmented dynamics, which is what the certificate establishes. Stability of the free equilibria is the separate statement of Section 6.1, $\operatorname{tr} A = -3$ and $\det A = 51/25 > 0$, and it is a property of $A = D_z F$ rather than of $D_z g$. Both facts are statements about a continuum of nudging strengths, not about the one value finally selected.

Nudging selection. The controller evaluates candidates in descending order and accepts the first one whose entire obligation list passes: tube certificate, both endpoint certificates, endpoint containment in the tube, and the interval error bound against the exact gradient of (28) below tolerance $\varepsilon = 10^{-5}$. Table 2 gives the outcome for a dense grid.

The selected strength is $\beta_{\max} = 0.005$, the largest tested candidate whose bound is below tolerance. This is not a claim of optimality: it is the largest accepted element of the tested set under the stated certificates. Table 3 reports the branch-level outcome at the selected strength.

Three properties of Table 3 should be recorded. The exact gradient lies outside both difference intervals, because the finite-nudge estimator is biased: the certificate is an enclosure of the estimator, and the bound then measures the estimator against the truth. The interval enclosure is tiny (width 4×10^{-8}) because the endpoint boxes were certified at radius 10^{-10}; the bound is dominated by

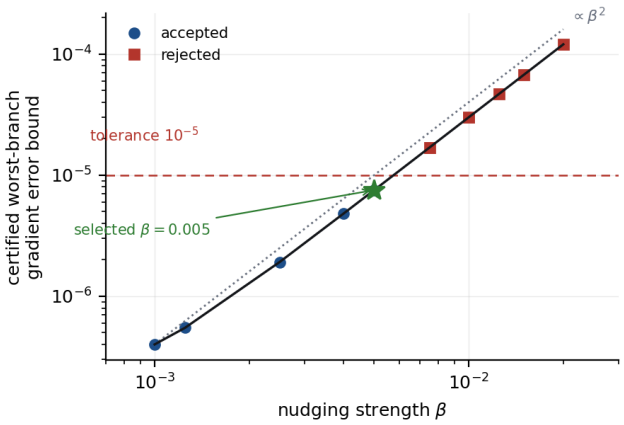


Figure 3: Certified worst-branch error bound (Table 2) against the nudging strength, log-log scale. Squares are rejected candidates, circles accepted ones, the star marks the selected $\beta_{\max} = 0.005$, and the dotted line is the quadratic reference. The vertical gap between the curve and the tolerance line is what the certificate contributes: the decision is made on the bound, not on an observed error.

the bias, not by root uncertainty. Finally, the observed floating-point error is close to but smaller than the bound, as it must be.

Rejection is part of the result. Two rejection paths are exercised in the same run. A deliberately wide box $[-1.2, 1.2] \times [-0.1, 0.1]$, which contains all three exact equilibria, fails the inclusion test with contraction factor $183/50 = 3.66 > 1$: the certificate refuses to report a unique root, which is the correct behaviour for a set containing three of them. Requests outside the certified parameter domain, for instance $\beta = 0.1$, raise a domain error rather than returning a weaker bound, so a tolerance cannot be met by silently leaving the certified region. Finally, an independent high-precision root computation (70 decimal digits) is verified to fall inside each certified endpoint box; it serves as a cross-check and carries no part of the guarantee, which comes from the interval inclusion.

6.2 A three-state non-normal benchmark

The second benchmark is a smooth, three-state, strongly non-normal system with a closed-form equilibrium branch,

$$F(z, \theta) = A_0\big(z - z^\star(\theta)\big) + \gamma \tanh\big(K(z - z^\star(\theta))\big), \qquad z^\star(\theta) = L\theta + o, \tag{31}$$

with

$$A_0 = \begin{pmatrix} -2 & 1.2 & -0.4 \\ -0.3 & -1.8 & 0.5 \\ 0.2 & -0.7 & -1.5 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 0.5 & 0.1 \\ -0.2 & 0 & 0.3 \\ 0.1 & -0.4 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0.2 \\ -0.4 & 0.7 \\ 0.3 & -0.5 \end{pmatrix}, \tag{32}$$

$\gamma = 0.25$, $o = (0.1, -0.2, 0.05)$, $\theta^\star = (0.2, -0.15)$, so that $z^\star = (0.27, -0.385, 0.185)$; the loss is $C(x) = \frac{1}{2}(x-t)^\top Q(x-t) + c\sum_i(1-\cos x_i)$ with $Q = \mathrm{diag}(1, 1.4, 0.8)$, $c = 0.1$ and $t = (0.5, -0.1, 0.3)$. The equilibrium Jacobian is $A = A_0 + \gamma K$ with eigenvalues $-1.8097 \pm 0.9749\,\mathrm{i}$ and -1.6805; despite being Hurwitz, A is far from normal, which makes this benchmark informative (Section 5 and Theorem 15).

Table 4: Certified budget on the three-state benchmark at $\beta = 0.005$ and integration horizon $T = 6.0$. The middle column records the value at the shorter horizon $T = 4.0$ that the controller first tried: the solve term then exceeds the tolerance, so the certificate forces a longer horizon rather than a relaxed tolerance. All quantities are produced by `certified_gep.py` in exact rational arithmetic where marked certified.

Term	$T = 4.0$	$T = 6.0$ (selected)
endpoint residual, + and −	3.12×10^{-6}, 3.26×10^{-6}	7.04×10^{-8}, 7.48×10^{-8}
distance to equilibrium, + and − (certified)	1.10×10^{-7}, 1.17×10^{-7}	1.10×10^{-7}, 1.17×10^{-7}
contrast bias E_β (certified)	3.656×10^{-4}	3.656×10^{-4}
solver term E_z (certified)	3.233×10^{-3}	7.351×10^{-5}
total certified bound	3.599×10^{-3}	4.391×10^{-4}
tolerance	5×10^{-4}	5×10^{-4}
verdict	rejected	accepted

Verification of the gradient identities. The implicit gradient of Theorem 2 agrees with a central finite difference of the objective to a relative error of 6.6×10^{-11}, and with reverse-mode automatic differentiation of the objective to 1.8×10^{-16}; the doubled-state stationary identity of Theorem 7 reproduces the same direction to 1.8×10^{-16}. The uncorrected vector-field rule (5) has relative error 1.55 on this benchmark, that is, it is not approximately right at all. The centred contrast of the corrected phase converges with observed log-log order 2.002 (one-sided: 0.946), as predicted by Theorem 9; the doubled-state signal is exact in algebra and limited only by the linear solve (Figure 1).

Certified residual run. The exact rational implementation certifies the following constants on the certified domain $|\beta| \le 0.32$: one-sided contraction rate $\alpha = 0.64$; parameter Jacobian bound $\|B\|_F \le 3.24$; $\|q(x_0)\| \le 0.488$; loss-gradient Lipschitz constant $L_q \le 1.5$; force derivative bounds $M_2 \le 0.066$, $M_3 \le 0.080$. None of these is fitted or assumed: each is an assertion in exact rational arithmetic that the code re-evaluates and that raises if violated. The contraction rate comes from a Gershgorin row-sum bound on the Jacobian of the learning field, computed over an exact interval enclosure of the state and compared with the claimed -0.64; the parameter-Jacobian bound from an exact Frobenius sum, checked against 3.24; the loss-gradient bound from an interval evaluation of q on the certified box; and the derivative bounds M_2, M_3 from the product structure of the force, with the row norms of K themselves bounded by rational constants. From these, the derivative bounds of Theorem 9 are computed by implicit differentiation and evaluated at $\beta = 0.005$ as

$$x_1 \le 0.7714, \qquad x_2 \le 3.6779, \qquad x_3 \le 27.0817, \qquad E_\beta = \frac{\|B\|\,\beta^2 x_3}{6} = 3.656 \times 10^{-4}. \tag{33}$$

The two perturbation endpoints are integrated with a high-accuracy solver and then re-evaluated in exact rational arithmetic: their residual infinities are bounded by 7.04×10^{-8} and 7.48×10^{-8}, which yields certified distances to the equilibrium of 1.10×10^{-7} and 1.17×10^{-7} and a solve contribution $E_z = \|B\|(r^+ + r^-)/(2\beta) = 3.24 \times (1.0999 \times 10^{-7} + 1.1688 \times 10^{-7})/(2 \times 0.005) = 7.351 \times 10^{-5}$, the enclosure term of (21) in its radius form (Table 4). The total certified bound is 4.391×10^{-4} at tolerance 5×10^{-4}, while the observed gradient error at the same settings is 1.076×10^{-5}: the certificate is 41 times more conservative than the observed error, and it bounds it.

Adaptive controller and integrator comparison. An independent adaptive controller selects (β, T, h) from the budget alone, without access to the true gradient: it seeds T from the exact matrix-

Table 5: Integrator comparison at the settings selected by the adaptive controller ($\beta = 0.04$, $T = 7.078$). DOP853 at tolerance 10^{-12} is used as a high-accuracy reference; its evaluation count is an internal output-interval count rather than a directly comparable right-hand-side call count, and it is therefore reported only to indicate order of magnitude.

Integrator	right-hand-side evaluations	accepted steps	gradient error vs reference
adaptive RK4 (step doubling)	168	14	7.88×10^{-7}
fixed RK4, $h = 0.05$	1136	284	2.57×10^{-11}
DOP853, tolerance 10^{-12}	8500	708	reference

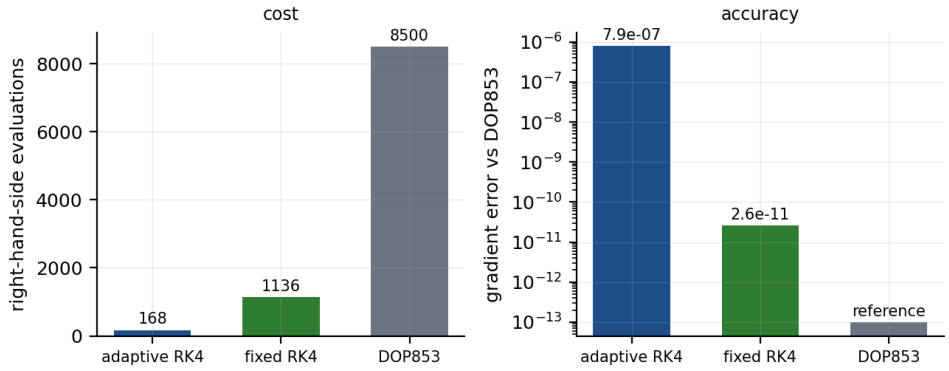


Figure 4: Cost and accuracy of the three integrators at the selected settings. The adaptive scheme reaches 7.9×10^{-7} gradient accuracy with 168 evaluations, two orders of magnitude fewer than the high-accuracy reference; the fixed-step scheme spends 1136 evaluations to gain four further digits. Neither number enters a certificate: they describe the floating-point tier of evidence.

exponential directional tail, extends it until both the tail and the observed finite-β change pass their thresholds, and scores candidate nudging strengths by a combined indicator. It selects $\beta = 0.04$, $T = 7.078$, with a combined indicator of 5.995×10^{-4} made up of a contrast estimate 1.932×10^{-4}, a finite-time change 2.251×10^{-5}, a numerical indicator 1.597×10^{-5} and a state-uncertainty bound 3.679×10^{-4} for a per-phase uncertainty budget of 5×10^{-6}. Its realised direction differs from the exact gradient by 1.938×10^{-4} in absolute terms (6.18×10^{-4} relative), and the adjacent-scale comparison gives an observed order of 2.0006, consistent with Theorem 9. The same run compares integrators at the selected settings (Table 5) and performs a 500-trial bounded state-noise stress test, in which the triangle inequality combining the noise-free error and the noise-induced term holds in every trial.

6.3 Baselines from the same run

Table 1 already compares the methods on the multistable benchmark. The doubled-state certificate of Theorem 8 is exercised in two ways. First, with an assumed residual bound $\|r_d\|_\infty \le 10^{-10}$ and the exact constant $\|A^{-\top}\|_\infty = 55/51$, the bound (16) evaluates to $11/10200000000 = 1.0784 \times 10^{-9}$ at $\beta = 0.1$ and to $11/10200000 = 1.0784 \times 10^{-6}$ at $\beta = 10^{-4}$, growing as $1/|\beta|$: the same amplification mechanism as in the budget. Second, and more stringently, the float64 linear solve that produces $\widehat{d}$ is re-evaluated in exact rational arithmetic: each returned coordinate is read as its exact binary rational value, the residual $r_d = A^\top \widehat{d} - \beta q$ is recomputed with rational arithmetic, and the resulting bound covers the observed error in all four cases (Table 6). This replaces the residual

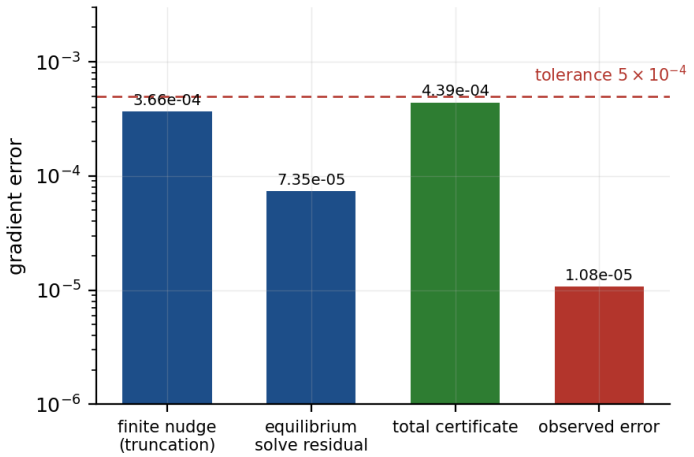


Figure 5: The certified budget of Table 4 at $T = 6.0$, with the observed error for comparison. The certificate is dominated by the contrast bias and is 41 times larger than the observed error, which is the intended direction: a bound tighter than the observed error would be an error, not a certificate.

Table 6: Exact rational re-evaluation of the float64 difference solve on the benchmark (25), branch $(1, 0)$. The residual is recomputed from the exact binary rational values of the returned coordinates; the bound is (16) with $\|A^{-\top}\|_\infty = 55/51$; the observed error is the distance of $\widehat{d}/\beta$ from the exact gradient $-11/34$.

β	exact residual $\|r_d\|_\infty$	certified bound	observed error
0.1	1.110×10^{-17}	1.1973×10^{-16}	5.7144×10^{-17}
0.01	9.541×10^{-19}	1.0289×10^{-16}	4.5919×10^{-17}
0.001	6.505×10^{-20}	7.0154×10^{-17}	3.5077×10^{-17}
0.0001	8.809×10^{-21}	9.5001×10^{-17}	4.3846×10^{-17}

from an assumption to a measurement, while remaining a statement about a linear subproblem with an exactly known midpoint.

6.4 Non-normality inflates finite-time bounds but not the error

The family $A_K = \begin{pmatrix} -1 & K \\ 0 & -0.1 \end{pmatrix}$ has fixed eigenvalues -1 and -0.1 and increasing non-normality measured by the logarithmic norm $\mu_2(A_K) = \frac{1}{2}\lambda_{\max}(A_K + A_K^\top)$ and by $\kappa_2(A_K)$. The first-order finite-time adjoint tail is $A^{-\top}e^{A^\top t}q$ with q chosen, adversarially, as the right singular vector of $A^{-\top}$ associated with its smallest singular value, so that the initial tail is small and can grow.

The smallest instance is a two-dimensional counterexample, which we state explicitly: $A = \begin{pmatrix} -1 & 10 \\ 0 & -1 \end{pmatrix}$ with $q = (0, 1)^\top$ gives $A^{-1}q = (-10, -1)$ and $A^{-\top}q = (0, -1)$, a relative gap of 10, while $\kappa_2(A) = 101.99$ and the semigroup norm peaks at 3.716 at $t = 0.98$. This instance belongs to the equal-eigenvalue family of Theorem 15 and is not the $K = 10$ row of Table 7, which has eigenvalues -1 and -0.1 and hence the larger condition number 1010.10. A finite-time argument based on the eigenvalues would predict monotone decay throughout and would therefore be in error by an order of magnitude in both the response and its bound.

Table 7: Transient behaviour of the adjoint response on the family A_K. The two eigenvalues are -1 and -0.1 for every row, the spectral abscissa is constant, and yet both the semigroup norm and the finite-time adjoint error grow by more than an order of magnitude. The last column is the ratio of the largest finite-time gradient error to its initial value.

K	$\kappa_2(A_K)$	μ_2	$\max_t \|e^{A_K^\top t}\|_2$	time of maximum	gradient error amplification
0	10.0	−0.100	1.000	0.00	1.000
1	20.05	0.123	1.125	1.66	1.000
2	50.08	0.547	1.738	2.30	1.518
5	260.10	1.990	3.949	2.51	3.857
10	1010.10	4.470	7.782	2.55	7.735
20	4010.10	9.460	15.505	2.56	15.481

6.5 Reproducibility

The package runs in under a minute on a laptop, with no dependence on precomputed artefacts. Running the driver executes, in order, the analytic validation, the bias and trade-off audits, the adaptive controller, the exact rational residual certificate, the multibranch Krawczyk certificate, the baseline comparison, the two doubled-state certificate stages and the dense nudging scan, and then the unit test suite; it writes logs and JSON artefacts and fails if any stage or any of the eight unit tests fails. The unit tests assert the properties that the paper relies on: two stable branches and one saddle with exact traces and determinants; passing tube certificates with contraction below one; the selection of $\beta = 1/200$ with a worst-branch bound below tolerance; containment of endpoint boxes in tubes and enclosure of the floating-point finite difference; rejection of the wide multibranch box; containment of 70-digit roots in the certified boxes; fail-closed behaviour outside the certified domain; and the exact gradient values $67/102$ and $-11/34$.

7 Discussion

What has been certified, and what has not. The analytic statements of Sections 4–5 are unconditional given Assumption 1: the gradient identity, the bias criterion, the sensitivity-adjoint dichotomy, the second-order contrast bound and the residual transfer inequality hold for any smooth non-conservative system with a regular equilibrium. The numerical certificates are statements about the two benchmarks, and their conclusions are intervals that contain the exact rational answer: on the multistable benchmark, both stable branches and both perturbation endpoints are certified, the parameter tube is certified uniformly over $\beta \in [-0.02, 0.02]$, stability holds throughout the tube, and the update is accepted or rejected against a tolerance by comparing a certified bound rather than an observed error. What is not claimed is any statement about trajectories that have not converged, any global count of equilibria, any performance comparison with backpropagation, or any transfer to systems whose coefficients are estimated from data.

Relation to finite-nudge exactness in thermal ensembles. For Gibbs ensembles the two-phase contrast is exactly the gradient of a free-energy difference at any finite nudging strength and temperature, and the deterministic rule is recovered in the zero-temperature limit [9]. Our setting is the deterministic one, and our object is the gap that remains there: the finite-nudge deficit of the deterministic contrast, bounded in Theorem 9 with a computable constant. The two results are complementary rather than competing. The thermal identity says that finite nudging is not a

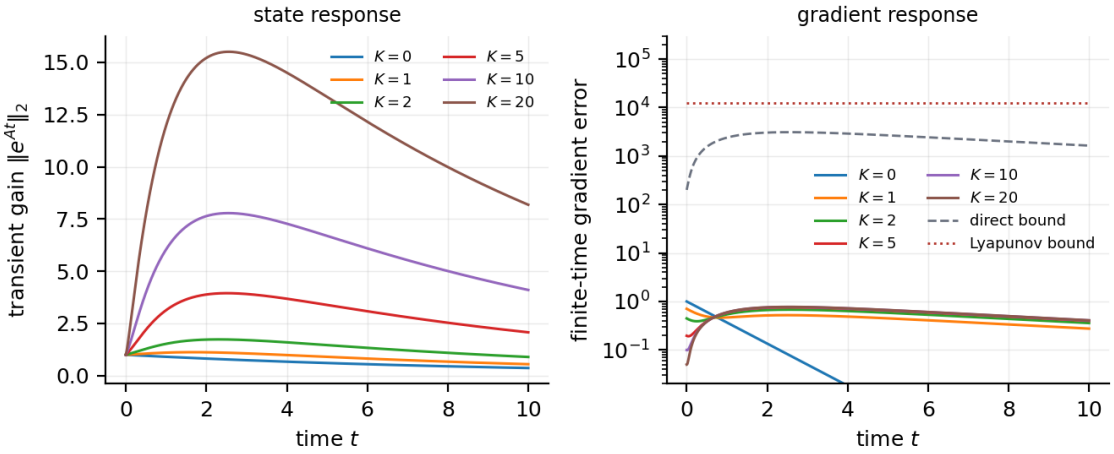


Figure 6: Left: semigroup norm $\|e^{A_K^\top t}\|_2$ for the family of Table 7; the eigenvalues are -1 and -0.1 in every curve. Right: the finite-time adjoint error $\|A^{-\top}e^{A^\top t}q\|$ (solid), the direct operator-norm bound (dashed) and the Lyapunov envelope (dotted) for $K = 20$, log scale. The direct bound is valid but three orders of magnitude loose; the Lyapunov envelope is four. Both are needed as options, because the direct bound is unavailable when only a Lyapunov function is known, and the envelope is unusable when sharpness matters.

bias when the ensemble is sampled correctly; the deterministic analysis says how large the bias is when it is not, and the certificate says when the resulting estimate may be trusted.

Relation to exact non-conservative algorithms. The corrected learning phase and the doubled-state construction are prior work [17], as is the compartmental construction behind dual propagation [4]; the bias of the uncorrected rule, and the separation of finite-nudge from asymmetry bias, are from [8, 16]. This paper does not claim any of the update rules. It contributes the conditions under which their computed output carries a guarantee, and the numerical machinery that produces one.

The two failure modes that certificates detect. Two failure modes escape residual-based reasoning. First, the sensitivity-adjoint gap of Proposition 3: on the benchmark the uncorrected rule is in error by 12% relative while its residual is 10^{-17}, so a small residual certifies nothing about the direction. Second, non-normality: Table 7 shows that eigenvalues fixed at -1 and -0.1 permit a 15.5-fold transient amplification of the adjoint response, so a finite-time error bound must come from a semigroup or integrator certificate and not from stability of the equilibrium. Both are properties of the model, and both are handled by the budget of Theorem 11: the first by measuring against the implicit gradient rather than against the estimator's own fixed point, the second by carrying a genuine finite-time term.

Cost. Certification requires interval evaluation of the Jacobian on a box, at cost $O(n^3)$ per inclusion test for a dense Jacobian, plus exact rational arithmetic whose integers grow with the number of operations. At the scale of the benchmarks ($n = 2$ and $n = 3$) the entire certified pipeline runs in well under a second, including both branches, both endpoints, the dense nudging grid and the wide-box rejection. The practical question for larger systems is not the asymptotic cost of interval arithmetic but whether the boxes stay tight enough for the inclusion to succeed; that is an empirical

question in a dimension not examined here, and it is the main direction for future work.

Limitations. (i) All statements are local: they concern a certified box around a regular isolated equilibrium, not global properties of the dynamics. (ii) The regularity requirement $\sigma_{\min}(A) > 0$ excludes bifurcation points and near-singular branches; the framework would have to be extended, not merely enlarged, to cover them. (iii) The benchmarks are of dimension two and three; cost and tightness in higher dimension are unmeasured. (iv) Coefficients, the loss and the free equilibrium are treated as exactly known rational data; measurement noise, parameter intervals and model mismatch are outside the certificate, and the state-noise stress test only *checks* a triangle bound rather than certifying it. (v) Certificates cover stationary states and finite-difference gradients, not validated envelopes of entire trajectories. (vi) The corrector $2A_{\mathrm{a}}$ is assumed exactly available, which in hardware would require an estimate; the certificate would then inherit the estimation error as an additional term. (vii) The proofs have been implemented as tests over exact rational arithmetic, but they have not been formalised in a proof assistant.

8 Conclusion

Equilibrium propagation beyond conservative systems is now supported by exact-gradient algorithms and by thermal-ensemble exactness results. The output of a run, however, is a pair of numerical states and a quotient by a small number. This paper has argued that the missing object is a certificate, and has supplied one: exact gradient conditions for smooth non-conservative systems with regular equilibria, a finite-perturbation error budget whose constants are computed rather than fitted, a branch-aware interval certificate that establishes existence and uniqueness uniformly over a parameter interval, and a fail-closed update rule that returns an interval or refuses. On an exactly solvable non-conservative system with two stable branches and a saddle, the pipeline certifies both branches, both perturbation endpoints and the whole nudging tube in rational arithmetic, accepts $\beta = 0.005$ with a worst-branch bound of 7.53×10^{-6} at tolerance 10^{-5}, and rejects both larger nudging strengths and a box that spans all three equilibria. The two benchmarks support one general observation: in non-conservative and non-normal systems, neither a small residual nor a stable spectrum implies that an EP update is what it appears to be, and only a certificate distinguishes the two cases.

A Proofs

Proof of Theorem 2. Differentiating $F(z^\star(\theta), \theta) = 0$ with respect to θ gives $A\, D_\theta z^\star + B = 0$, hence $D_\theta z^\star = -A^{-1}B$. Since $L(\theta) = C(z^\star(\theta), \theta)$, the chain rule gives $\nabla_\theta L = \partial_\theta C + (D_\theta z^\star)^\top q = \partial_\theta C - B^\top A^{-\top} q$. □

Proof of Proposition 3. Let $\varphi(\beta)$ denote the root of the uncorrected nudged equation (5) near $z^\star$. Differentiating $F(\varphi(\beta)) - \beta \nabla C(\varphi(\beta)) = 0$ at $\beta = 0$ gives $A\varphi'(0) - q = 0$, that is $\varphi'(0) = A^{-1}q = s$. The centred difference $(\varphi(\beta) - \varphi(-\beta))/(2\beta)$ equals $\varphi'(0)$ up to the even-order term $\frac{1}{6}\varphi'''(0)\beta^2$. The parameter gradient requires $\lambda = A^{-\top}q$ by Theorem 2, and $A^{-1} = A^{-\top}$ exactly when $A = A^\top$. □

Proof of Theorem 4. Combining the infinitesimal response $A^{-1}q$ for (5) with the contrast (4) gives $\widehat{g}_0 = B^\top A^{-1} q$, while the true negative gradient is $B^\top A^{-\top} q$; subtracting yields (11). Exactness for a fixed q is therefore equivalent to $B^\top (A^{-1} - A^{-\top})q = 0$, and exactness for all q in a subspace S is equivalent to (12). The claim of Example 5 follows by direct evaluation of the 2×2 matrices. □

Proof of Theorem 6. With $g(z,\beta) = F(z) - 2A_{\mathrm{a}}(z - z^\star) - \beta\nabla C(z)$ we have $D_z g(z^\star, 0) = A - 2A_{\mathrm{a}} = A - (A - A^\top) = A^\top$. Differentiating $g(\varphi(\beta), \beta) = 0$ at $\beta = 0$ gives $A^\top \varphi'(0) - \nabla C(z^\star) = 0$, hence $\varphi'(0) = A^{-\top} q = \lambda$. □

Proof of Theorem 7. By hypothesis the midpoint satisfies $F(m) = 0$ and the difference satisfies $A(m)^\top d = \beta\nabla C(m)$; multiplying by $A(m)^{-\top}$ (which exists because $A(m)$ is nonsingular) gives $d = \beta A(m)^{-\top}\nabla C(m)$ for every $\beta \neq 0$. If m is the free equilibrium and C does not depend explicitly on θ, then $B^\top d/\beta = B^\top A^{-\top} q$, which is the true negative gradient of Theorem 2. □

Proof of Theorem 8. Substituting the definition of r_d into $\widehat{d}/\beta - A(\widehat{m})^{-\top}\nabla C(\widehat{m})$ gives $\beta^{-1}A(\widehat{m})^{-\top}\big(A(\widehat{m})^\top \widehat{d} - \beta\nabla C(\widehat{m})\big) = \beta^{-1}A(\widehat{m})^{-\top} r_d$, and taking norms yields (16). The constant $55/51$ is the infinity norm of $\begin{pmatrix} -25/51 & 5/51 \\ -5/51 & -50/51 \end{pmatrix}$, the exact inverse transpose of A at the benchmark branch, verifiable in rational arithmetic. □

Proof of Theorem 9. The map $\beta \mapsto \varphi(\beta)$ is C^3 in a neighbourhood of $\beta = 0$ by the implicit function theorem applied to $(z,\beta) \mapsto g(z,\beta)$ at $(z^\star, 0)$, whose Jacobian $A^\top$ is nonsingular by Assumption 1. Write the third-order expansion $\varphi(\beta) = z^\star + \beta\lambda + \tfrac{1}{2}\beta^2\varphi''(0) + R(\beta)$ with $\|R(\beta)\| \le \tfrac{1}{6}M_3|\beta|^3$. Then $(\varphi(\beta) - \varphi(-\beta))/(2\beta) = \lambda + (R(\beta) - R(-\beta))/(2\beta)$, whose norm is at most $\tfrac{1}{6}M_3\beta^2$ by the triangle inequality and the bound on R. Applying $\widehat{g}_\beta - d^\star = B^\top\big((\varphi(\beta) - \varphi(-\beta))/(2\beta) - \lambda\big)$ yields (18). □

Proof of Theorem 12. Let $\psi(z) = z - P\,g(z)$. For $z \in X$ the mean value theorem gives $\psi(z) = x - P\,g(x) + \big(I - P\,Dg(\xi)\big)(z - x)$ for some ξ on the segment from x to z. Since $Dg(\xi)$ lies in the interval enclosure $Dg(X)$, the right-hand side lies in $K(X)$, so ψ maps X into $\mathrm{int}(X)$ whenever the inclusion holds. In addition $\|D\psi(X)\|_\infty = \|I - P\,Dg(X)\|_\infty < 1$, so ψ is a contraction on X and has exactly one fixed point there by the Banach fixed-point theorem; fixed points of ψ coincide with zeros of g because P is nonsingular. The parametric statement of Corollary 13 follows by applying the argument for each fixed β with the same box, which the interval evaluation over $\mathcal{B}$ covers uniformly. □

Proof of Theorem 15. For $A_K = \begin{pmatrix} -1 & K \\ 0 & -1 \end{pmatrix}$ we have $A_K^\top = \begin{pmatrix} -1 & 0 \\ K & -1 \end{pmatrix}$ and $e^{A_K^\top t} = e^{-t}\begin{pmatrix} 1 & 0 \\ Kt & 1 \end{pmatrix}$, so $\|e^{A_K^\top t}\|_2^2 = e^{-2t}\lambda_{\max}\big(\begin{pmatrix} 1 + K^2t^2 & Kt \\ Kt & 1 \end{pmatrix}\big)$, which yields (24) after evaluating the larger eigenvalue. The Lyapunov inequality $\|e^{A^\top t}\|_2 \le \sqrt{\kappa_2(P)}e^{-\lambda t}$ with $AP + PA^\top = -I$ and $\lambda = 1/(2\|P\|_2)$ is standard; its looseness on the scanned family is measured in Table 7 and Figure 6. □

B Exact rational constants

The following quantities appear in the certificates and are computed in rational arithmetic; the second column gives the exact fraction, the third its decimal form.

Quantity	Exact value	Decimal
true gradient, branch $(-1, 0)$	$67/102$	0.6568627451
true gradient, branch $(1, 0)$	$-11/34$	-0.3235294118
determinant of A at ± 1	$51/25$	2.04
trace of A at ± 1	-3	-3
preconditioned contraction, both tubes	$2249/13600$	0.1653676471
trace interval of the learning Jacobian	$[-1339/400, -1067/400]$	$[-3.3475, -2.6675]$
determinant interval of the learning Jacobian	$[4219/2500, 24199/10000]$	$[1.6876, 2.4199]$
contraction of the rejected wide box	$183/50$	3.66
infinity norm of $A^{-\top}$ on the benchmark	$55/51$	1.0784313725
tolerance	$1/100000$	10^{-5}

C Reproducibility

The package contains nine runnable stages and one test module: `run_validation.py` (analytic gradient, finite differences, error decomposition), `run_stage2.py` (task-restricted exactness, nudging trade-off, non-normal transient), `run_stage3.py` (adaptive controller, integrator comparison, tighter non-normal bounds), `certified_gep.py` (exact rational residual certificate on the three-state benchmark), `certified_multibranch_gep.py` (exact rational multibranch Krawczyk certificate), `beta_scan.py` (dense nudging scan and the data of Figure 3), `unified_baselines.py`, `dyadic_residual_certificate.py` and `dyadic_finite_precision_certificate.py` (baselines and the two doubled-state certificates), plus `tests/test_stage5.py` with eight assertions. `run_all.py` executes all of them, writes `results/` and `logs/`, runs the test suite, and exits non-zero on any failure; `make_figures.py` regenerates every figure of this paper from the artefacts. The certificate JSON files carry both decimal and exact fractional forms of every accepted bound, so a reader can re-verify an accepted claim without re-running the pipeline.

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